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Electronic device and antenna module

Abstract

An electronic device includes a metal cover and an antenna module. The metal cover includes two cover slots. The antenna module includes a substrate, an antenna structure, and a bottom structure. Both sides of the substrate are provided with the antenna structure and the bottom structure that are connected. A plurality of projection regions respectively defined by orthogonally projecting a first excitation segment and a second excitation segment of the antenna structure onto the metal cover are partially overlapped with the two cover slots. The first excitation segment and the second excitation segment are connected to each other. The bottom structure has two antenna slots. When a signal source is fed into the antenna structure through a feeding portion, the antenna structure, the two antenna slots, and the two cover slots can cooperate with each other to generate three frequency bands with different frequency ranges.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

(1) This application claims the benefit of priority to Taiwan Patent Application No. 112143630, filed on Nov. 13, 2023. The entire content of the above identified application is incorporated herein by reference.

(2) Some references, which may include patents, patent applications and various publications, may be cited and discussed in the description of this disclosure. The citation and/or discussion of such references is provided merely to clarify the description of the present disclosure and is not an admission that any such reference is “prior art” to the disclosure described herein. All references cited and discussed in this specification are incorporated herein by reference in their entireties and to the same extent as if each reference was individually incorporated by reference.

FIELD OF THE DISCLOSURE

(3) The present disclosure relates to an electronic device and an antenna module, and more particularly to an electronic device for three different frequency bands and an antenna module for three different frequency bands.

BACKGROUND OF THE DISCLOSURE

(4) A conventional electronic device (e.g., a smartphone, a tablet computer, or a laptop computer) is

provided with a plurality of antennas therein by manufacturers for wireless communication in different frequency bands. Accordingly, the conventional electronic device can be operated for wireless communication in different frequency bands through different antennas.

(5) In practice, since the interior space in the conventional electronic device (e.g., the smartphone, the tablet computer, or the laptop computer) for installing the antennas is relatively small, manufacturers may encounter an issue where the plurality of antennas cannot all be installed in the conventional electronic device. Therefore, how a necessary number of the antennas can be installed in an electronic device having a smaller interior space has become an issue for relevant manufacturers.

SUMMARY OF THE DISCLOSURE

(6) In response to the above-referenced technical inadequacies, the present disclosure provides an electronic device and an antenna module for improving on the issues associated with conventional electronic devices (e.g., smartphones or laptop computers). Specifically, the issues are described as follows. In order to receive wireless signals in a variety of different frequency bands, the conventional electronic device needs to be installed with a plurality of antennas. As overall appearances of smartphones, laptop computers and other devices trend toward thinner and lighter, multiple antennas capable of receiving different frequency bands are difficult to be installed into such devices in order to maintain the reception efficiency of the antennas.

(7) In order to solve the above-mentioned problems, one of the technical aspects adopted by the present disclosure is to provide an electronic device. The electronic device includes a metal cover and an antenna module. The metal cover has two cover slots arranged side by side and penetrating therethrough, and lengths of the two cover slots are different from each other. The antenna module includes a substrate, an antenna structure, and a bottom structure. The substrate is disposed on one side of the metal cover. The substrate has a first side surface and a second side surface that is opposite to the first side surface. The antenna structure is a conductive structure and is formed on the first side surface. The antenna structure defines a feeding portion, and the antenna structure includes a first excitation segment, a second excitation segment, and a connecting segment. A projection region defined by orthogonally projecting the first excitation segment onto the metal cover is at least partially overlapped with one of the two cover slots. A projection region defined by orthogonally projecting the second excitation segment onto the metal cover is at least partially overlapped with another one of the two cover slots. The connecting segment that connects the first excitation segment and the second excitation segment. The bottom structure is electrically conductive and is disposed on the second side surface, the bottom structure is connected to the antenna structure, and the bottom structure is grounded through the substrate. The bottom structure has two antenna slots, and lengths of the two antenna slots are different from each other. When a signal source is fed into the antenna structure through the feeding portion, the antenna structure, one of the two antenna slots, and the one of the two cover slots are configured to generate a first frequency band, the antenna structure, another one of the two antenna slots, and the another one of the two cover slots are configured to generate a second frequency band, and the antenna structure is configured to generate a third frequency band. Moreover, a frequency range of the first frequency band, a frequency range of the second frequency band, and a frequency range of the third frequency band are not completely identical to each other.

(8) In order to solve the above-mentioned problems, another one of the technical aspects adopted by the present disclosure is to provide an antenna module. The antenna module is fixed on a metal cover, the metal cover includes a metal cladding layer structure and a non-metallic inner layer structure. An outer surface of the metal cover being arranged on the metal cladding layer structure and the metal cladding layer structure envelopes the non-metal inner structure. The metal cover has two cover slots arranged side by side and penetrating therethrough, lengths of the two cover slots are different from each other, and the antenna module includes a substrate, an antenna structure, and a bottom structure. The substrate is disposed on one side of the metal cover, and the substrate

has a first side surface and a second side surface that is opposite to the first side surface. The antenna structure is a conductive structure and is formed on the first side surface. The antenna structure defines a feeding portion, and the antenna structure includes a connecting segment, a first excitation segment, and a second excitation segment. The first excitation segment extends from one end of the connecting segment in an extending direction. The second excitation segment extends from another one end of the connecting segment in the extending direction. The bottom structure is conductive and is disposed on the second side surface, the bottom structure is connected to the antenna structure, the bottom structure is grounded through the substrate, the bottom structure has two antenna slots, and lengths of the two antenna slots are different from each other. When the antenna module is fixed on the metal cover, a projection region defined by orthogonally projecting the first excitation segment onto the metal cover is at least partially overlapped with one of the two cover slots, and a projection region defined by orthogonally projecting the second excitation segment onto the metal cover is at least partially overlapped with another one of the two cover slots. A signal source is fed into the antenna structure through the feeding portion, the antenna structure, one of the two antenna slots, the one of the two cover slots are configured to generate a first frequency band, the antenna structure, another one of the two antenna slots, the another one of the two cover slots are configured to generate a second frequency band, and the antenna structure is configured to generate a third frequency band. A frequency range of the first frequency band, a frequency range of the second frequency band, and a frequency range of the third frequency band are not completely identical to each other.

(9) Therefore, in the electronic device and antenna module provided by the present disclosure, the first excitation segment, the second excitation segment, and the connecting segment of the antenna structure are designed to cooperate with the long cover slot and the short cover slot, such that when the signal source is fed into the antenna structure through the feeding portion, the antenna module can generate three different frequency bands. In addition, the design of the bottom structure allows the frequency range of the antenna for different bands to be easily adjusted according to the material composition of the metal cover (e.g., the metal cover being made of pure metal, or the metal cover being mixed with non-metallic materials in addition to metal materials).

(10) These and other aspects of the present disclosure will become apparent from the following description of the embodiment taken in conjunction with the following drawings and their captions, although variations and modifications therein may be affected without departing from the spirit and scope of the novel concepts of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The described embodiments may be better understood by reference to the following description and the accompanying drawings, in which:

(2) FIG. 1 is a partial schematic view of an electronic device according to a first embodiment of the present disclosure;

(3) FIG. 2 is a partial schematic exploded view of the electronic device according to the first embodiment of the present disclosure;

(4) FIG. 3 is a partial schematic exploded view of the electronic device according to the first embodiment of the present disclosure;

(5) FIG. 4 is a schematic cross-sectional view taken along line IV-IV of FIG. 1;

(6) FIG. 5 is a partial schematic side view of the electronic device according to the first embodiment of the present disclosure;

(7) FIG. 6 is a voltage standing wave ratio (VSWR) chart of the electronic device according to the first embodiment of the present disclosure;

- (8) FIG. 7 is a radiation efficiency chart of the electronic device according to the first embodiment of the present disclosure;
- (9) FIG. 8 is a schematic view showing a bottom structure of an antenna module according to a second embodiment of the present disclosure;
- (10) FIG. 9 is a partial schematic side view of the electronic device according to the second embodiment of the present disclosure;
- (11) FIG. 10 is a partial schematic side view of the electronic device according to a third embodiment of the present disclosure;
- (12) FIG. 11 is a partial schematic side view of the electronic device according to a fourth embodiment of the present disclosure;
- (13) FIG. 12 is a partial schematic side view of the electronic device according to a fifth embodiment of the present disclosure;
- (14) FIG. 13 is a partial schematic view of the electronic device according to a sixth embodiment of the present disclosure;
- (15) FIG. 14 is a partial schematic view of the electronic device according to a seventh embodiment of the present disclosure;
- (16) FIG. 15 is a partial schematic exploded view of the electronic device according to the seventh embodiment of the present disclosure; and
- (17) FIG. 16 is a schematic view of the electronic device according to an eighth embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(18) The present disclosure is more particularly described in the following examples that are intended as illustrative only since numerous modifications and variations therein will be apparent to those skilled in the art. Like numbers in the drawings indicate like components throughout the views. As used in the description herein and throughout the claims that follow, unless the context clearly dictates otherwise, the meaning of “a,” “an” and “the” includes plural reference, and the meaning of “in” includes “in” and “on.” Titles or subtitles can be used herein for the convenience of a reader, which shall have no influence on the scope of the present disclosure.

(19) The terms used herein generally have their ordinary meanings in the art. In the case of conflict, the present document, including any definitions given herein, will prevail. The same thing can be expressed in more than one way. Alternative language and synonyms can be used for any term(s) discussed herein, and no special significance is to be placed upon whether a term is elaborated or discussed herein. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms is illustrative only, and in no way limits the scope and meaning of the present disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given herein.

Numbering terms such as “first,” “second” or “third” can be used to describe various components, signals or the like, which are for distinguishing one component/signal from another one only, and are not intended to, nor should be construed to impose any substantive limitations on the components, signals or the like.

(20) Referring to FIG. 1 to FIG. 4, FIG. 1 is a partial schematic view of an electronic device according to a first embodiment of the present disclosure; FIG. 2 is a partial schematic exploded view of the electronic device according to the first embodiment of the present disclosure; FIG. 3 is another partial schematic exploded view of the electronic device according to the first embodiment of the present disclosure; and FIG. 4 is a schematic cross-sectional view taken along line IV-IV of FIG. 1.

(21) The present disclosure provides an electronic device **100**, which includes a metal cover **1** and an antenna module **2**. The electronic device **100** can be a smartphone, a tablet computer, or a laptop computer, but the present disclosure is not limited thereto. The metal cover **1** can be an upper cover or a back cover of the electronic device **100**, but the present disclosure is not limited thereto.

(22) The metal cover **1** has two cover slots. The two cover slots are respectively defined as a short cover slot **11** and a long cover slot **12**. The short cover slot **11** penetrates through the metal cover **1**, and the long cover slot **12** penetrates through the metal cover **1**. In practice, any one of the short cover slot **11** and the long cover slot **12** can be a rectangular hole having an enclosed contour, but the shape of the short cover slot **11** and the shape of the long cover slot **12** are not limited thereto. A length **11W** of the short cover slot **11** is less than a length **12W** of the long cover slot **12**. Moreover, a difference between the length **11W** of the short cover slot **11** and the length **12W** of the long cover slot hole **12** can be changed according to requirements, and the present disclosure is not limited thereto.

(23) In the present embodiment, the short cover slot **11** and the long cover slot **12** are arranged side by side, and a short side of the short cover slot **11** and a short side of the long cover slot **12** are arranged adjacent to each other. However, the short cover slot **11** and the long cover slot **12** are not limited to being arranged side by side.

(24) The antenna module **2** includes a substrate **21**, an antenna structure **22**, a grounding member **23**, and a bottom structure **27**. The substrate **21** is disposed on one side of the metal cover **1**. The antenna structure **22** is a conductive structure, and the antenna structure **22** is formed on a side surface of the substrate **21**. In practice, the substrate **21** can be a flame retardant 4 (FR4) substrate, a printed circuit board (PCB), or a flexible circuit board (FCB), according to requirements, and the present disclosure is not limited thereto. The antenna structure **22** can be a conductive sheet structure (e.g., a copper foil) that is formed on the side surface of the substrate **21**. Two opposite width-side surfaces of the substrate **21** are a first side surface **211** and a second side surface **212**, respectively. The antenna structure **22** is disposed on the first side surface **211**, and the bottom structure **27** is electrically conductive and is disposed on the second side surface **212**.

(25) The antenna structure **22** defines a feeding portion **221**. The feeding portion **221** is configured to connect a signal source, such that signals transmitted from the signal source are fed into the antenna structure **22** through the feeding portion **221**. The antenna structure **22** includes a first excitation segment **222**, a second excitation segment **223**, and a connecting segment **224**. Two ends of the connecting segment **224** are respectively connected to the first excitation segment **222** and the second excitation segment **223**. The first excitation segment **222**, the second excitation segment **223**, and the connecting segment **224** are integrally formed on the substrate **21**. The first excitation segment **222** extends from one end of the connecting segment **224** in an extending direction, and the second excitation segment **223** extends from another end of the connecting segment **224** in the extending direction. The first excitation segment **222**, the second excitation segment **223**, and the connecting segment **224** are jointly formed as an inverted U-shaped structure. The feeding portion **221** is arranged on the first excitation segment **222**.

(26) In a specific embodiment, the grounding member **23** is a conductive structure, and the grounding member **23** is provided for grounding. The grounding member **23** includes a first grounding structure **231**, a grounding sheet body **232**, and a second grounding structure **233**. The first grounding structure **231** and the second grounding structure **233** are formed on the first side surface **211** of the substrate **21**. The first grounding structure **231** and the second grounding structure **233** are arranged adjacent to the first excitation segment **222** and the second excitation segment **223**, respectively. Each of the first grounding structure **231** and the second grounding structure **233** can be a rectangular sheet. In practice, the second grounding structure **233** can be connected to an end of the second excitation segment **223**, and the second grounding structure **233** and the second excitation segment **223** can be integrally formed as a single one-piece structure.

(27) The grounding sheet body **232** is coupled to the first grounding structure **231** and the second excitation segment **223**. In practice, the first grounding structure **231** and the grounding sheet body **232** can be integrally connected to each other, or the grounding structure **231** and the grounding sheet body **232** can be fixed to each other in a soldering manner through solders, but the present disclosure is not limited thereto. In practice, a part of the grounding sheet body **232** can be fixed to

the metal cover **1** through various manners.

(28) As shown in FIG. **3**, the bottom structure **27** is a conductive structure and the bottom structure **27** is connected to the antenna structure **22**. The bottom structure **27** is grounded through the substrate **21**. The bottom structure **27** includes two antenna slots, and lengths of the two antenna slots are different from each other. The two antenna slots are respectively defined as a long antenna slot **271** and a short antenna slot **272**, the long antenna slot **271** and the short antenna slot **272** are respectively two enclosed holes (e.g., two rectangular holes respectively having an enclosed contour) that are independent from each other. An overall shape of the bottom structure **27** is formed approximately in the shape of a figure-8 that defines the two antenna slots.

(29) As shown in FIG. **4**, in one of specific embodiments, the substrate **21** can include a plurality of through-holes **213** penetrating therethrough. Each of the through-holes **213** is fully filled with one of a plurality of conductive structures **214**. The through-holes **213** respectively correspond in position to the first grounding structure **231** and the second grounding structure **233**, and the antenna structure **22** is connected to the bottom structure **27** through the conductive structures **214**. The through-holes **213** and the conductive structures **214** can be formed together during the manufacturing process of the substrate **21**, but the present disclosure is not limited thereto. Moreover, the connection manner between the bottom structure **27** and the antenna structure **22** is not limited by the present embodiment, and the bottom structure **27** and the antenna structure **22** can be connected to each other in other manners.

(30) The forming manner and the sequence of forming for the first grounding structure **231**, the second grounding structure **233**, the antenna structure **22**, the bottom structure **27**, the through-holes **213**, and the conductive structures **214** can be designed according to practical requirements, but the present disclosure is not limited thereto. For example, in one of embodiments, after the antenna structure **22**, the bottom structure **27**, and the ground member **23** are formed on the substrate **21**, the through-holes **213** are formed in the first grounding structure **231** and the second grounding structure **233**, and then the conductive structures **214** are formed in the through-holes **213**.

(31) It should be noted that, in different embodiments, the grounding member **23** can be in a configuration where the first grounding structure **231** is not included, the grounding sheet body **232** is directly connected to an end of the first excitation section **222**, and the grounding sheet body **232** is directly connected to the bottom structure **27** through the conductive structures **214**.

(32) In practice, a cover intermediate structure **15** is arranged between the two cover slots, and the bottom structure **27** has an antenna intermediate structure **275** that is arranged between the two cover slots. In practice, a first length **15W** of the cover intermediate structure **15** can be approximately equal to a second length **275W** of the antenna intermediate structure **275**. A length **12W** of the long cover slot **12** is greater than a length **271W** of the long antenna slot **271**, and a length **11W** of the short cover slot **11** is greater than a length **272W** of the short antenna slot **272**.

(33) In addition, a height **271H** of the long antenna slot **271** can be approximately equal to a height **272H** of the short antenna slot **272**. A height **12H** of the long cover slot **12** can be approximately equal to a height **11H** of the short cover slot **11**. In practice, the height **271H** of the long antenna slot **271**, the height **272H** of the short antenna slot **272**, the height **12H** of the long cover slot **12**, and the height **11H** of the short cover slot **11** can be approximately the same, but the present disclosure is not limited thereto. In different embodiments, the height **271H** of the long antenna slot **271** and the height **272H** of the short antenna slot **272** are be slightly greater than the height **12H** of the long cover slot **12** and the height **11H** of the short cover slot **11**, respectively.

(34) As shown in FIGS. **1** and **4**, it should be noted that, in one of embodiments, the metal cover **1** can include a metal cladding layer structure **16** and a non-metallic inner layer structure **17**. An outer surface of the metal cover **1** is arranged on the metal cladding layer structure **16**, and the metal cladding layer structure **16** envelops the non-metallic inner structure **17**. In other words, the non-metallic inner structure **17** is embedded in the metal cladding layer structure **16**. For example, the

metal cladding layer structure **16** is substantially made of aluminum-magnesium alloy material, and the non-metallic inner structure **17** is substantially made of plastic material. That is to say, the metal cover **1** can be a structure having an aluminum “skin” and a plastic “skeleton” that is covered by the aluminum skin. In a special embodiment, the non-metallic inner structure **17** of the metal cover **1** can be incompletely enveloped by the metal cladding layer structure **16**, and a part of the non-metallic inner structure **17** is exposed from the metal cladding layer structure **16**.

(35) As shown in FIG. 1 to FIG. 5, the antenna module **2** is fixed on one side of the metal cover **1** that is formed with the short cover slot **11** and the long cover slot **12**. Specifically, the bottom structure **27** faces toward the metal cover **1**, the long antenna slot **271** faces toward the long cover slot **12**, the short antenna slot **272** faces toward the short cover slot **11**, and the antenna intermediate structure **275** substantially corresponds in position to the cover intermediate structure **15**. A projection region defined by orthogonally projecting the first excitation segment **222** onto the metal cover **1** is at least partially overlapped with the short cover slot **11**, and is at least partially overlapped with the short antenna slot **272**. A projection region defined by orthogonally projecting the second excitation segment **223** onto the metal cover **1** is at least partially overlapped with the long cover slot **12**, and is at least partially overlapped with the long antenna slot **271**. In other words, as shown in the partial schematic side view of FIG. 5, the first excitation segment **222** of the antenna structure **22** is arranged across the short cover slot **11**, and the second excitation segment **223** is arranged across the long cover slot **12**.

(36) It should be noted that, as shown in FIG. 5 of the present embodiment, an upper edge **224T** of the connecting segment **224** is not overlapped with an upper edge **12T** of the long cover slot **12**, an upper edge **11T** of the short cover slot **11**, an upper edge **271T** of the long antenna structure **271**, and an upper edge **272T** of the short antenna slot **272**, but the present disclosure is not limited thereto. In different embodiments, the upper edge **224T** of the connecting segment **224** can be overlapped with the upper edge **12T** of the long cover slot **12**, the upper edge **11T** of the short cover slot **11**, the upper edge **271T** of the long antenna slot **271**, and the upper edge **272T** of the short antenna slot **272**.

(37) In summary, when a signal source is fed into the antenna structure **22** through the feeding portion **221**, the antenna structure **22** is configured to interact with the short cover slot **11** and the short antenna slot **272** so as to generate a first frequency band, the antenna structure **22** is configured to interact with the long cover slot **12** and the long antenna slot **271** so as to generate a second frequency band, and the antenna structure **22** is configured to be resonated so as to generate a third frequency band. Moreover, frequency ranges respectively corresponding to the first frequency band, the second frequency band, and the third frequency band are not entirely identical to each other. Specifically, the frequency ranges respectively corresponding to the first frequency band, the second frequency band, and the third frequency band can be completely different from each other; or, the frequency ranges respectively corresponding to the first frequency band and the second frequency band can be partially overlapped with each other, and the frequency ranges respectively corresponding to the second frequency band and the third frequency band are partially overlapped with each other. In the embodiment where each of the long cover slot **12** and the short cover slot **11** is a rectangular hole having the enclosed contour, the length **11W** of the short cover slot **11** can be within a range from $\frac{1}{2}$ times to 4 times of a wavelength corresponding to a center frequency of the first frequency band, and the length **12W** of the long cover slot **12** can be within a range from $\frac{1}{2}$ times to 4 times of a wavelength of a center frequency of the second frequency band.

(38) The electronic device **100** of the present disclosure further includes a coaxial cable **3**. The coaxial cable **3** includes an internal conductor **31** and an external conductor **32**, and the internal conductor **31** and the external conductor **32** are electrically isolated from each other through an insulation structure. The external conductor **32** is coupled to the first grounding structure **231**, and the internal conductor **31** is coupled to the feeding portion **221** of the antenna structure **22**. The external conductor **32** can be connected to the grounding sheet body **232** in a soldering manner

through solders, and the external conductor **32** is electrically connected to the first grounding structure **231** through the grounding sheet body **232**. The internal conductor **31** can be connected and fixed to the feeding portion **221** of the antenna structure **22** in a soldering manner through solders. However, the external conductor **32** of the coaxial cable **3** can be directly connected to the first grounding structure **231** without using the grounding sheet body **232** so as to achieve the same connection effect. It is worth mentioning that, the grounding sheet body **232** can further include a notch **2321**. The notch **2321** corresponds in position to the feed portion **221**, and the notch **2321** is used to prevent the positive pole and negative pole of the coaxial cable **3** from being connected to each other. In practice, a width of the notch **2321** is provided according to a position of the feed portion **221** and a position of the first grounding structure **231**, but the present disclosure is not limited thereto. In a preferred embodiment, a distance between the feed portion **221** of the antenna structure **22** and the grounding member **23** that is connected to the coaxial cable **3** is not less than 2 mm.

(39) Specifically, the electronic device **100** and the antenna module **2** provided by a specific embodiment of the present disclosure, a relevant personnel can change the frequency ranges respectively corresponding to the first frequency band, the second frequency band, and the third frequency bands by adjusting a position of the first excitation segment **222** relative to the short cover slot **11**, adjusting a position of the second excitation segment **223** relative to the long cover slot **12**, adjusting a position of the first excitation segment **222** relative to the short antenna slot **272**, adjusting a position of the second excitation segment **223** relative to the long antenna slot **271**, changing a shape of the first excitation segment **222**, changing a shape of the second excitation segment **223**, or changing an overall dimensions or a shape of the antenna structure **22**.

(40) Please refer to FIG. **6**, which shows a voltage standing wave ratio (VSWR) chart of the electronic device **100** according to the first embodiment of the present disclosure. As shown in FIG. **4**, after the signal source is fed into the electronic device **100** of the present disclosure, a voltage standing wave ratio of the electronic device **100** at 2.4 GHz is 1.8592, the voltage standing wave ratio of the electronic device **100** at 2.5 GHz is 1.8921, the voltage standing wave ratio of the electronic device **100** at 5.15 GHz is 1.2116, the voltage standing wave ratio of the electronic device **100** at 5.85 GHz is 1.8933, the voltage standing wave ratio of the electronic device **100** at 6.125 GHz is 1.8054, and the voltage standing wave ratio of the electronic device **100** at 7.125 GHz is 1.3170. Therefore, the electronic device **100** and the antenna module **2** of the present disclosure can be provided to support a common broadband operations at 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz and 5 GHz.

(41) Please refer to FIG. **7**, which is a radiation efficiency chart of the electronic device **100** according to the first embodiment of the present disclosure. As shown in FIG. **7**, the antenna module **2** of the present disclosure operated in any one of three frequency ranges of 6000-7125 MHz, 2310-2600 MHz and 5150-5850 MHz has a radiation efficiency of more than 20%. Therefore, the antenna module **2** of the present disclosure already meets the application requirements of common frequencies such as 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz, and 5 GHz. It should be noted that, in the examples shown in FIG. **6** and FIG. **7**, the metal cover **1** includes the metal cladding layer structure **16** and the non-metallic inner layer structure **17**.

(42) Please refer to FIG. **3** again, in order to enable the antenna structure **22** of the electronic device **100** to generate three different frequency bands of 6000-7125 MHz, 2310~2600 MHz, and 5150~5850 MHz in applications where the metal cover **1** includes the metal cladding layer structure **16** and the non-metallic inner layer structure **17**, a difference between the length **15W** of the cover intermediate structure **15** and the second length **275W** of the antenna intermediate structure **275** is less than 40% of the first length **15W**, the length **271W** of the long antenna slot **271** is greater than or equal to 80% of the length **12W** of the long cover slot **12**, and the length **272W** of the short antenna slot **272** is greater than or equal to 80% of the length **11W** of the short cover slot **11**.

(43) Please refer to FIG. **5** again, in order to enable the antenna structure **22** of the electronic device

100 to generate three different frequency bands of 6000-7125 MHz, 2310~2600 MHz, and 5150~5850 MHz, the short cover slot **11** in practice is the rectangular hole having the enclosed contour, a shortest distance **D1** between the first excitation segment **222** and one side of the short cover slot **11** is less than or equal to 10 mm, the long cover slot **12** is the rectangular hole having the enclosed contour, and a shortest distance **D2** between the second excitation segment **223** and one side of the long cover slot **12** is less than or equal to 10 mm.

(44) Please refer to FIG. 2 to FIG. 4 again, it should be noted that, in each figure of present embodiment, a length **21W** of the substrate **21** is greater than a sum of the first length **15W** of the cover intermediate structure **15**, the length **11W** of the short cover slot **11**, and the length **12W** of the long cover slot hole **12** (i.e., $21W \geq 15W + 11W + 12W$). When the substrate **21** is fixed onto the metal cover **1**, the substrate **21** correspondingly covers the long cover slot **12** and the short cover slot **11**. However, the present disclosure is not limited to require the substrate **21** completely covering the long cover slot **12** and the short cover slot **11** when the substrate **21** is fixed onto one side of the metal cover **1**. In other words, when the substrate **21** is fixed onto one side of the metal cover **1**, as the conditions required by the present disclosure are described as follows: the projection region defined by orthogonally projecting the first excitation segment **222** onto the metal cover **1** is overlapped with the short cover slot **11**, the projection region defined by orthogonally projecting the second excitation segment **223** onto the metal cover **1** is overlapped with the long cover slot **12**, a projection region defined by orthogonally projecting the first excitation segment **222** onto the second side surface **212** of the substrate **21** is overlapped with the short antenna slot **272**, and a projection region defined by orthogonally projecting the second excitation segment **223** onto the second side surface **212** of the substrate **21** is overlapped with the long antenna slot **271**.

(45) It should be noted that, when the electronic device **100** and the antenna module **2** are provided without the bottom structure **27** and are applied with the metal cover **1** having mixed non-metallic materials (e.g., the metal cover **1** having the aluminum “skin” and the plastic “skeleton”), the electronic device **100** and the antenna module **2** are difficult to meet the application requirements of 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz and 5 GHz. In other words, no matter how the relevant personnel modifies the shape or the structure of the antenna module **2**, the electronic device **100** and the antenna module **2** are difficult to achieve the frequency ranges of 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz and 5 GHz. Conversely, the electronic device **100** and the antenna module **2** of the present disclosure can effectively meet the common application requirements of 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz and 5 GHz through a cooperation design of the antenna structure **22**, the bottom structure **27** and the grounding member **23**. In other words, if the metal cover **1** includes the metal cladding layer structure **16** and the non-metallic inner layer structure **17**, the electronic device **100** and the antenna module **2** of the present disclosure can still meet the application requirements of 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz, and 5 GHz by simply adjusting the antenna structure **22** and the bottom structure **27**.

(46) Specifically, a dielectric coefficient of the metal cover **1** only made of metal materials (e.g., the metal cover **1** having an aluminum skin and magnesium ribs) is completely different from that of the metal cover **1** made of metal materials mixed with non-metallic materials (e.g., the metal cover **1** having the aluminum “skin” and the plastic “skeleton”). When the metal cover **1** is mixed with non-metallic materials, the dielectric coefficient of the metal cover **1** can be changed according to the thickness of the non-metallic materials. Therefore, when the antenna module **2** is provided without the bottom structure **27** and is applied to the metal cover **1** that is only made of metal materials, and the antenna module **2** reaches the frequency ranges that complies with 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz and 5 GHz, the antenna module **2** can be directly applied with the metal cover **1** mixed with non-metal materials (e.g., the metal cover **1** with the aluminum “skin” and the plastic “skeleton”), and the antenna module **2** cannot be able to produce the same frequency band due to the change in the dielectric coefficient of the metal cover **1**. That is to say, the antenna module **2** have a frequency offset issue, such that the antenna module **2** can not reach the frequency

ranges of 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz and 5 GHz.

(47) Please refer to FIG. 8 and FIG. 9, FIG. 8 is a schematic view showing the bottom structure of the antenna module according to a second embodiment of the present disclosure, and FIG. 9 is a partial schematic side view of the electronic device according to the second embodiment of the present disclosure. The main difference between the present embodiment and the previous embodiment is described as follows: the bottom structure 27 of the present embodiment further includes two frequency-adjustment structures. The two frequency-adjustment structures are respectively a first bottom frequency-adjustment structure 273 and a second bottom frequency-adjustment structure 274. The first bottom frequency-adjustment structure 273 is formed by extending from an inner side edge of the long antenna slot 271 toward an opposite inner side edge of the long antenna slot 271, and an overall shape of the bottom frequency-adjustment structure 273 protrudes from a rectangular structure of the long antenna slot 271. A height 273H of the first bottom frequency-adjustment structure 273 is less than or equal to the height 271H of the long antenna slot 271. The second bottom frequency-adjustment structure 274 is formed by recessing in an inner side edge of the short antenna slot 272 along a direction away from an opposite inner side edge of the short antenna slot 272, and an overall shape of the second bottom frequency-adjustment structure 274 is recessed in a notch of the short antenna slot 272.

(48) As shown in FIG. 9, after the antenna module 2 is fixed on the metal cover 1, a projection region defined by orthogonally projecting the first bottom frequency-adjustment structure 273 onto the metal cover 1 is not overlapped with the long cover slot 12, and a projection region defined by orthogonally projecting the second bottom frequency-adjustment structure 274 onto the metal cover 1 is not overlapped with the short cover slot 11.

(49) In summary, the relevant personnel can adjust at least one of the shape of the first bottom frequency-adjustment structure 273, a length 273W of the first bottom frequency-adjustment structure 273, a position of the first bottom frequency-adjustment structure 273 in the long antenna slot 271, and a height 273H of the first bottom frequency-adjustment structure 273 so as to modify a low-frequency band (e.g., 2310 MHz-2600 MHz) generated by the antenna module 2 and the metal cover 1. Similarly, the relevant personnel can adjust at least one of a shape of the second bottom frequency-adjustment structure 274, a length 274W of the second bottom frequency-adjustment structure 274, a position of the second bottom frequency-adjustment structure 274 in the long antenna slot 271, and a height 274H of the second bottom frequency-adjustment structure 274 so as to modify a 5G band (e.g., 5150 MHz-5850 MHz) generated by the antenna module 2 and the metal cover 1. In other words, the relevant personnel can adjust the first bottom frequency-adjustment structure 273 and the second bottom frequency-adjustment structure 274 so as to correspondingly modify the frequency ranges of the antenna module 2 at 2.4 GHz and 5 GHz.

(50) Please refer to FIG. 10, which shows a partial schematic side view of the electronic device according to a third embodiment of the present disclosure. A difference between the present embodiment and the first embodiment is described as follows: the antenna structure 22 of the present embodiment further includes two frequency-adjustment segments. The two frequency-adjustment segments are respectively defined as a first frequency-adjustment segment 225 and a second frequency-adjustment segment 226. The first frequency-adjustment segment 225 is connected to the connecting segment 224 and the first excitation segment 222, and a projection region defined by orthogonally projecting the first frequency-adjustment segment 225 onto the metal cover 1 is at least partially overlapped with the short cover slot 11. In addition, the relevant personnel can adjust at least one of a shape and a size of the first frequency-adjustment segment 225 and a position of the first frequency-adjustment segment 225 with respect to the short cover slot 11 so as to modify the frequency range of the first frequency band generated by the antenna module 2 when the signal source is fed into the antenna module 2. In other words, the first frequency-adjustment segment 225 is mainly provided to modify the frequency range of the antenna module 2 in 6 GHz (e.g., Wi-Fi 6E).

(51) The second frequency-adjustment segment **226** is connected to the connecting segment **224** and the second excitation segment **223**, and a projection region defined by orthogonally projecting the second frequency-adjustment segment **226** onto the metal cover **1** is at least partially overlapped with the long cover slot **12**. In addition, the relevant personnel can adjust at least one of a shape and a size of the second frequency-adjustment segment **226** and a position of the second frequency-adjustment segment **226** with respect to the long cover slot **12** so as to modify the frequency range of the second frequency band generated by the antenna module **2** when the signal source is fed into the antenna module **2**. In other words, the second frequency-adjustment segment **226** is mainly provided to modify the frequency range of the antenna module **2** at 2.4 GHz. Regarding the drawings of present embodiment, the shapes of the first frequency-adjustment segment **225** and the second frequency-adjustment segment **226** can be designed according to requirements, but the present disclosure is not limited thereto.

(52) Please refer to FIG. **11**, which shows a partial schematic side view of the electronic device according to a fourth embodiment of the present disclosure. A difference between the present embodiment and the first embodiment is described as follows: a projection region defined by orthogonally projecting the connecting segment **224** of the antenna structure **22** onto the metal cover **1** (as show in FIG. **2**) can be not overlapped with the long cover slot **12**, the short cover slot **11**, the long antenna slot **271**, and the short antenna slot **272**. Furthermore, the antenna structure **22** further includes the first frequency-adjustment segment **225**. Regarding the description of the first frequency-adjustment segment **225**, please refer to the third embodiment, and the present embodiment is not described herein for the sake of brevity. The shape and size of the first frequency-adjustment segment **225** can be changed according to the frequency band generated by the antenna module **2** and the metal cover **1**, and are not limited by present embodiment or the third embodiment.

(53) Please refer to FIG. **12**, which shows a partial schematic side view of the electronic device according to a fifth embodiment of the present disclosure. A difference between the present embodiment and the previous embodiment is described as follows: the antenna structure **22** of the present embodiment further includes two frequency-adjustment supports. The two frequency-adjustment supports are respectively defined as a first frequency-adjustment support **24** and a second frequency-adjustment support **25**. The first frequency-adjustment support **24** is connected to the first grounding structure **231**, and the first frequency-adjustment support **24** is coupled to the grounding sheet body **232**. The second frequency-adjustment support **25** is connected to the second grounding structure **233**, and the second frequency-adjustment support **25** is coupled to the grounding sheet body **232**.

(54) A projection region defined by orthogonally projecting the first frequency-adjustment support **24** onto the metal cover **1** is at least partially overlapped with the short cover slot **11** and the short antenna slot **272**, and a projection region defined by orthogonally projecting the second frequency-adjustment support **25** onto the metal cover **1** is at least partially overlapped with the long cover slot **12** and the long antenna slot **271**. In practice, the first frequency-adjustment support **24** and the first grounding structure **231** can be integrally formed as a single one-piece structure, and the second frequency-adjustment support **25** and the second grounding structure **233** can be integrally formed as a single one-piece structure. In different embodiments, the first grounding structure **231**, the grounding sheet body **232**, the second grounding structure **233**, the first frequency-adjustment support **24**, and the second frequency-adjustment support **25** can be integrally formed as a single one-piece structure. In practice, the relevant personnel can adjust at least one of a shape and a size of the first frequency-adjustment support **24** and a position of the first frequency-adjustment support **24** with respect to the short cover slot **11** and the short antenna slot **272** so as to modify the frequency range corresponding to the first frequency band generated by a cooperation of the antenna structure **22**, the short cover slot **11**, and the short antenna slot **272** when the signal source is fed into the antenna structure **22**. In addition, the relevant personnel can adjust at least one of a

shape and a size of the second frequency-adjustment support **25** and a position of the second frequency-adjustment support **25** with respect to the long cover slot **12** and the long antenna slot **271** so as to modify the frequency range corresponding to the second frequency band generated by a cooperation of the antenna structure **22**, the long cover slot **12**, and the long antenna slot **271** when the signal source is fed into the antenna structure **22**.

(55) Please refer to FIG. **13**, which shows a partial schematic view of the electronic device according to a sixth embodiment of the present disclosure. A maximum difference between the present embodiment and the previous embodiment is described as follows: the feeding portion **221** is arranged on an end of the second excitation segment **223** that is opposite to another end of the second excitation segment **223** connected to the connecting segment **224**, and the feeding portion **221** is correspondingly arranged adjacent to the long cover slot **12**. Another difference between the present embodiment and the previous embodiment is described as follows: the feeding portion **221** is disposed on the first excitation segment **222**.

(56) In practice, if the feeding portion **221** is disposed on the first excitation segment **222**, the antenna structure **22** is configured to enhance performance in the second frequency band (e.g., the efficiency of the antenna structure **22** in the second frequency band being enhanced). Similarly, if the feeding portion **221** is disposed on the second excitation segment **223**, the antenna structure **22** is configured to enhance performance in the first frequency band. More specifically, the feeding portion **221** can be provided to enhance the performance of the antenna module **2** at 2.4 GHz by being disposed on the first excitation segment **222**. The feeding portion **221** can be provided to enhance the performance of the antenna module **2** at a frequency band of 6 GHz (e.g., Wi-Fi 6E) by being disposed on the second excitation segment **223**.

(57) Another difference between the present embodiment and the previous embodiment is described as follows: the first frequency-adjustment segment **225** is arranged on a connecting position of the first excitation segment **222** and the connecting segment **224**, the second frequency-adjustment segment **226** and the second excitation segment **223** are connected to each other, and the second frequency-adjustment segment **226** is arranged on a connecting position of the second excitation segment **223** and the connecting segment **224**. More specifically, the first frequency-adjustment segment **225** can be a rectangular structure, a short side of the first frequency-adjustment segment **225** and the connecting segment **224** are connected to each other, and a long side of the first frequency-adjustment segment **225** and the first excitation segment **222** are connected to each other. The second frequency-adjustment segment **226** can be a rectangular structure, and a short side of the second frequency-adjustment segment **226** and the connecting segment **224** are connected to each other.

(58) In summary, in a specific embodiment, the relevant personnel can adjust shapes, sizes and positions of the first frequency-adjustment segment **225** and the second frequency-adjustment segment **226** so as to slightly modify the frequency range of the first frequency band, the frequency range of the second frequency band, and frequency range of the third frequency band that are generated by the antenna module **2**. That is to say, the shapes, the sizes, and the positions of the first frequency-adjustment segment **225** and the second frequency-adjustment segment **226** shown in the figure of the present embodiment are just one of implementation modes, and the actual application is not limited thereto.

(59) Please refer to FIG. **14** and FIG. **15**, FIG. **14** is a partial schematic view of the electronic device according to a seventh embodiment of the present disclosure, and FIG. **15** is a partial schematic exploded view of the electronic device according to the seventh embodiment of the present disclosure. A difference between the present embodiment and the first embodiment is described as follows: the antenna module **2** of the present embodiment further includes an auxiliary frequency-adjustment member **26**. The auxiliary frequency-adjustment member **26** is disposed on a side surface of the substrate **21**, and the antenna structure **22** is arranged between the auxiliary frequency-adjustment member **26** and the substrate **21**. The auxiliary frequency-adjustment

member **26** is a dielectric material. For example, the auxiliary frequency-adjustment member **26** can be a component composed of polymer, ceramic or other composite materials. When the auxiliary frequency-adjustment member **26** is fixed on the substrate **21**, the auxiliary frequency-adjustment member **26** correspondingly shields at least part of the first excitation segment **222**, at least part of the second excitation segment **223**, and at least part of the connecting segment **224**. (60) In practice, after the relevant personnel can select the auxiliary frequency-adjustment member **26** having different dielectric constants for adjusting the signal source to feed into the antenna module **2**, the antenna module **2** generates the frequency range corresponding to the first frequency band, the frequency range corresponding to the second frequency band, and the frequency range corresponding to the third frequency band. In addition, in practice, the substrate **21** can be fixed to the metal cover **1** through screws, and the auxiliary frequency-adjustment member **26** can include a threaded hole that is not cooperated with the screws, so that the screws do not need to be threaded to the auxiliary frequency-adjustment member **26**. It should be noted that, the auxiliary frequency-adjustment member **26** of the present embodiment can be combined with any one of the second to sixth embodiments so as to jointly form a new embodiment.

(61) Please refer to FIG. 1, FIG. 2, and FIG. 16, FIG. 16 is a schematic view of the electronic device according to an eighth embodiment of the present disclosure. As shown FIG. 1 and FIG. 2, in the aforementioned embodiments, the short cover slot **11** and the long cover slot **12** are formed on a wide side wall **14** of the metal cover **1**, and the substrate **21** is disposed on the wide side wall **14** of the metal cover **1**. In the present embodiment, the short cover slot **11** and the long cover slot **12** are formed on a narrow side wall **13** of the metal cover **1**, and the substrate **21** is disposed on the narrow side wall **13** of the metal cover **1**. A part of the grounding sheet body **232** of the grounding member **23** can be disposed on the wide side wall **14** of the metal cover **1**. In other words, as long as the short cover slot **11** and the long cover slot **12** can be cooperated with the antenna module **2** to generate the frequency band of 6 GHz (e.g., Wi-Fi 6E), the frequency band of 2.4 GHz, and the frequency band of 5 GHz, the short cover slot **11** and the long cover slot **12** can be disposed on any position of the metal cover **1**.

BENEFICIAL EFFECTS OF THE EMBODIMENTS

(62) In conclusion, in the electronic device and the antenna module provided by the present disclosure, after the signal source is fed into the antenna structure, the antenna module can generate three different frequency bands, thereby being applied to the electronic device having a small size or a small installation space. In addition, the electronic device and the antenna module of the present disclosure can be provided to generate the frequency bands of 6 GHz (e.g., Wi-Fi 6E), 2.4 GHz, and 5 GHz through the design of the bottom structure when the metal cover includes the metal cladding layer structure and the non-metallic inner layer structure.

(63) The foregoing description of the exemplary embodiments of the disclosure has been presented only for the purposes of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Many modifications and variations are possible in light of the above teaching.

(64) The embodiments were chosen and described in order to explain the principles of the disclosure and their practical application so as to enable others skilled in the art to utilize the disclosure and various embodiments and with various modifications as are suited to the particular use contemplated. Alternative embodiments will become apparent to those skilled in the art to which the present disclosure pertains without departing from its spirit and scope.

Claims

1. An electronic device, comprising: a metal cover having two cover slots arranged side by side and penetrating therethrough, wherein lengths of the two cover slots are different from each other; and an antenna module including: a substrate disposed on one side of the metal cover, wherein the

substrate has a first side surface and a second side surface that is opposite to the first side surface; an antenna structure being a conductive structure and formed on the first side surface, wherein the antenna structure defines a feeding portion, and the antenna structure includes: a first excitation segment, wherein a projection region defined by orthogonally projecting the first excitation segment onto the metal cover is at least partially overlapped with one of the two cover slots; a second excitation segment, wherein a projection region defined by orthogonally projecting the second excitation segment onto the metal cover is at least partially overlapped with another one of the two cover slots; and a connecting segment that connects the first excitation segment and the second excitation segment; and a bottom structure being electrically conductive and disposed on the second side surface, wherein the bottom structure is connected to the antenna structure, and the bottom structure is grounded through the substrate, and wherein the bottom structure has two antenna slots, and lengths of the two antenna slots are different from each other; wherein, when a signal source is fed into the antenna structure through the feeding portion, the antenna structure, one of the two antenna slots, and the one of the two cover slots are configured to generate a first frequency band, the antenna structure, another one of the two antenna slots, and the another one of the two cover slots are configured to generate a second frequency band, and the antenna structure is configured to generate a third frequency band, and wherein a frequency range of the first frequency band, a frequency range of the second frequency band, and a frequency range of the third frequency band are not completely identical to each other.

2. The electronic device according to claim 1, wherein the metal cover includes a metal cladding layer structure and a non-metallic inner layer structure, and wherein an outer surface of the metal cover is arranged on the metal cladding layer structure, and the metal cladding layer structure envelops the non-metal inner structure.

3. The electronic device according to claim 2, wherein the frequency range of the first frequency band is within a range from 6000 MHz to 7125 MHz, the frequency range of the second frequency band is within a range from 2310 MHz to 2600 MHz, and the frequency range of the third frequency band is within a range from 5150 MHz to 5850 MHz.

4. The electronic device according to claim 1, wherein the two cover slots are a short cover slot and a long cover slot, and a length of the short cover slot is less than a length of the long cover slot, wherein the two antenna slots are a short antenna slot and a long antenna slot, the short antenna slot faces toward the short cover slot, and the long antenna slot faces toward the long cover slot, and wherein the length of the short antenna slot is greater than or equal to 80% of the length of the short cover slot, and the length of the long antenna slot is greater than or equal to 80% of the length of the long cover slot.

5. The electronic device according to claim 4, wherein the short cover slot is a rectangular hole, and the length of the short cover slot is within a range from $\frac{1}{2}$ times to 4 times of a wavelength corresponding to a center frequency of the first frequency band, wherein the long cover slot is a rectangular hole, and the length of the long cover slot is within a range from $\frac{1}{2}$ times to 4 times of the wavelength of the center frequency of the first frequency band, and wherein each of the short antenna slot and the long antenna slot is a rectangular hole.

6. The electronic device according to claim 4, wherein the metal cover includes a cover intermediate structure arranged between the two cover slots, and the antenna module includes an antenna intermediate structure arranged between the two cover slots, wherein a difference between a length of the cover intermediate structure and a length of the antenna intermediate structure is less than 40% of the length of the cover intermediate structure.

7. The electronic device according to claim 1, wherein the two cover slots are a short cover slot and a long cover slot, and a length of the short cover slot is less than a length of the long cover slot, wherein the two antenna slots are a short antenna slot and a long antenna slot, the short antenna slot faces toward the short cover slot, and the long antenna slot faces toward the long cover slot, and wherein a width of the short antenna slot is greater than or equal to 80% of a width of the short

cover slot, a width of the long antenna slot is greater than or equal to 80% of a width of the long cover slot.

8. The electronic device according to claim 1, wherein the antenna module further includes at least one frequency-adjustment structure connected to the bottom structure, wherein the at least one frequency-adjustment structure is formed by extending from an inner side edge of one of the two antenna slots towards an opposite inner side edge of the one of the two antenna slots, or the at least one frequency-adjustment structure is formed by recessing the inner side edge of one of the two antenna slots towards a direction away from the opposite inner side edge of the one of the two antenna slots, and wherein a length of the at least one frequency-adjustment structure is less than a length of the one of the two antenna slots, and a height of the at least one frequency-adjustment structure is less than a height of the one of the two antenna slots.

9. The electronic device according to claim 1, wherein the antenna module further includes a grounding member, and the bottom structure is connected to the grounding member through the substrate, and wherein the grounding member is connected to one of the first excitation segment and the second excitation segment, and the feeding portion is arranged on another one of the first excitation segment and the second excitation segment.

10. The electronic device according to claim 9, wherein the substrate has a plurality of through-holes penetrating therethrough and arranged adjacent to a portion of the grounding member that is connected to the antenna structure, and each of the through-holes is fully filled with one of a plurality of conductive structures, and wherein the grounding member, the antenna structure, and the bottom structure are electrically coupled to each other through the conductive structures in the through-holes.

11. An antenna module for being fixed on a metal cover, the metal cover including a metal cladding layer structure and a non-metallic inner layer structure, an outer surface of the metal cover being arranged on the metal cladding layer structure, the metal cladding layer structure enveloping the non-metal inner structure, the metal cover having two cover slots arranged side by side and penetrating therethrough, lengths of the two cover slots being different from each other, and the antenna module comprising: a substrate disposed on one side of the metal cover, wherein the substrate has a first side surface and a second side surface that is opposite to the first side surface; and an antenna structure being a conductive structure and formed on the first side surface, wherein the antenna structure defines a feeding portion, and the antenna structure includes: a connecting segment; a first excitation segment extending from one end of the connecting segment in an extending direction; and a second excitation segment extending from another one end of the connecting segment in the extending direction; a bottom structure being electrically conductive and disposed on the second side surface, wherein the bottom structure is connected to the antenna structure, and the bottom structure is grounded through the substrate, and wherein the bottom structure has two antenna slots, and lengths of the two antenna slots are different from each other; wherein, when the antenna module is fixed on the metal cover, a projection region defined by orthogonally projecting the first excitation segment onto the metal cover is at least partially overlapped with one of the two cover slots, a projection region defined by orthogonally projecting the second excitation segment onto the metal cover is at least partially overlapped with another one of the two cover slots, wherein a signal source is fed into the antenna structure through the feeding portion, the antenna structure, one of the two antenna slots, and the one of the two cover slots are configured to generate a first frequency band, the antenna structure, another one of the two antenna slots, and the another one of the two cover slots are configured to generate a second frequency band, and the antenna structure is configured to generate a third frequency band, and wherein a frequency range of the first frequency band, a frequency range of the second frequency band, and a frequency range of the third frequency band are not completely identical to each other.

12. The antenna module according to claim 11, wherein the frequency range of the first frequency band is within a range from 6000 MHz to 7125 MHz, the frequency range of the second frequency

band is with a range from 2310 MHz to 2600 MHz, and the frequency range of the third frequency band is within a range from 5150 MHz to 5850 MHz.

13. The antenna module according to claim 11, wherein the two cover slots are a short cover slot and a long cover slot, a length of the short cover slot is less than a length of the long cover slot, wherein the two antenna slots are a short antenna slot and a long antenna slot, the short antenna slot faces toward the short cover slot, and the long antenna slot faces toward the long cover slot, and wherein the length of the short antenna slot is greater than or equal to 80% of the length of the short cover slot, and the length of the long antenna slot is greater than or equal to 80% of the length of the long cover slot.

14. The antenna module according to claim 13, wherein the short cover slot is a rectangular hole, and the length of the short cover slot is within a range from $\frac{1}{2}$ times to 4 times of wavelength corresponding to a center frequency of the first frequency band, wherein the long cover slot is a rectangular hole, and the length of the long cover slot is within a range from $\frac{1}{2}$ times to 4 times of wavelength corresponding to the center frequency of the first frequency band, and wherein each of the short antenna slot and the long antenna slot is a rectangular hole.

15. The antenna module according to claim 13, wherein the metal cover includes a cover intermediate structure arranged between the two cover slots, and the antenna module includes an antenna intermediate structure arranged between the two cover slots, wherein a difference between a length of the cover intermediate structure and a length the antenna intermediate structure is less than 40% of the length of the cover intermediate structure.

16. The antenna module according to claim 11, wherein the two cover slots are a short cover slot and a long cover slot, and a length of the short cover slot is less than a length of the long cover slot, wherein the two antenna slots are a short antenna slot and a long antenna slot, and the short antenna slot faces toward the short cover slot, the long antenna slot faces toward the long cover slot, and wherein a width of the short antenna slot is greater than or equal to 80% of a width of the short cover slot, and a width of the long antenna slot is greater than or equal to 80% of a width of the long cover slot.

17. The antenna module according to claim 11, wherein the antenna module further includes at least one frequency-adjustment structure connected to the bottom structure, wherein the at least one frequency-adjustment structure is formed by extending from an inner side edge of one of the two antenna slots towards an opposite inner side edge of the one of the two antenna slots, or the at least one frequency-adjustment structure is formed by recessing the inner side edge of one of the two antenna slots towards a direction away from the opposite inner side edge of the one of the two antenna slots, and wherein a length of the at least one frequency-adjustment structure is less than a length of the one of the two antenna slots, and a height of the at least one frequency-adjustment structure is less than a height of the one of the two antenna slots.

18. The antenna module according to claim 11, wherein the antenna module further includes a grounding member, and the bottom structure is connected to the grounding member through the substrate, and wherein the grounding member is connected to one of the first excitation segment and the second excitation segment, and the feeding portion is arranged on another one of the first excitation segment and the second excitation segment.

19. The antenna module according to claim 18, wherein the substrate has a plurality of through-holes penetrating therethrough and arranged adjacent to a portion of the grounding member that is connected to the antenna structure, and each of the through-holes is fully filled with one of a plurality of conductive structures, and wherein the grounding member, the antenna structure, and the bottom structure are electrically coupled to each other through the conductive structures in the through-holes.
